METRIC
MIL-DTL-23679G(NAVY)
14 June 2021
SUPERSEDING
DOD-R-23679F(SH)
8 August 1986

DETAIL SPECIFICATION

REAGENTS, STANDARD, WATER TEST CHEMICALS

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reagents for use in checking the quality of water testing chemicals and equipment used in boiler powered ships.

1.2 Classification. Standard reagents are of the following classes, as specified (see 6.2).

a. Class E – Standard hardness solution, 0.20 equivalent per million (epm) hardness.

b. Class I – Standard conductivity solution, 1400 microsiemens per centimeter (µS/cm).

c. Class K – Standard chloride solution, 2.00 epm chloride.

d. Class P – Standard methyl purple alkalinity solution, 6.0 epm.

1.3 Part or identifying number (PIN). PINs to be used for reagents acquired to this specification are created as follows:

M	**23679**	-	**X**
Prefix for Military Specification	Specification Number	-	Class (see 1.2)
Examples: M23679-E, M23679-P			

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-15000 - Water-Testing Chemicals, Boiler, Shipboard Use

MIL-DTL-24617 - Hardness Testing Reagents, Boiler Feedwater, Shipboard Use

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN CHEMICAL SOCIETY (ACS)

ACS Reagent Chemicals: Specifications and Procedures for Reagents and Standard-Grade Reference Materials

(Copies of this document are available online at http://pubs.acs.org/.)

ASTM INTERNATIONAL

ASTM D512 - Standard Test Methods for Chloride Ion in Water

ASTM D1125 - Standard Test Methods for Electrical Conductivity and Resistivity of Water

ASTM D1126 - Standard Test Method for Hardness in Water

ASTM D1193 - Standard Specification for Reagent Water

ASTM E287 - Standard Specification for Laboratory Glass Graduated Burets

ASTM E288 - Standard Specification for Laboratory Glass Volumetric Flasks

ASTM E969 - Standard Specification for Glass Volumetric (Transfer) Pipets

(Copies of these documents are available online at www.astm.org.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Chemical material. Materials used in the manufacture of each of the chemicals covered by this specification shall be in accordance with ACS Reagent Chemicals: Specifications and Procedures for Reagents and Standard-Grade Reference Materials.

3.1.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.1.2 Prohibited materials. The reagents shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.1.3 Composition. The material, as received, shall conform to the chemical and weight characteristics of table I when tested as specified in therein.

TABLE I. Chemical and weight characteristics.

Class	Description	Requirements Paragraph	Weight Requirement (Minimum)	Chemical Requirement (Range)	Test Paragraphs
E	Hardness	3.1.4	150 grams	0.20±0.03 epm hardness	4.2.1, 4.2.2
I	Conductivity	3.1.5	250 grams	1400±25 µS/cm conductivity	4.2.1, 4.2.3
K	Chloride	3.1.6	75 grams	2.00±0.04 epm chloride	4.2.1, 4.2.4
P	Alkalinity	3.1.7	100 grams	6.0±0.2 epm alkalinity	4.2.1, 4.2.5

3.1.4 Class E – standard hardness solution, 0.20 epm hardness. The class E standard hardness solution shall contain 0.0111 gram of reagent grade anhydrous calcium chloride per liter of type II reagent grade water and shall conform to the characteristics in table I when tested as specified therein.

3.1.5 Class I – standard conductivity solution, 1400 µS/cm. The class I standard conductivity solution shall contain 0.7440 gram of reagent grade anhydrous potassium chloride per liter of type II reagent grade water and shall conform to the characteristics in table I when tested as specified therein.

3.1.6 Class K – standard chloride solution, 2.00 epm chloride. The class K standard chloride solution shall contain 0.1170 gram of reagent grade anhydrous sodium chloride per liter of type II reagent water solution and shall conform to the characteristics in table I when tested as specified therein.

3.1.7 Class P – standard methyl purple alkalinity solution, 6.0 epm. The class P standard methyl purple alkalinity solution shall contain 0.318 gram of reagent grade anhydrous sodium carbonate per liter of type II reagent water solution and shall conform to the characteristics in table I when tested as specified therein.

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the samples specified in table II for the tests of 4.2.1 through 4.2.5.

TABLE II. Required samples for Government acceptance testing.

Class	Description	Samples Per Lot	Test Paragraphs
E	Hardness	6 bottles in one box	4.2.1, 4.2.2
I	Conductivity	6 bottles in one box	4.2.1, 4.2.3
K	Chloride	6 bottles in one box	4.2.1, 4.2.4
P	Alkalinity	6 bottles in one box	4.2.1, 4.2.5

4.1.1 Lot. A lot shall consist of all solutions or chemicals from one batch. In cases in which it is not practicable to define the sampling lot in this manner, the lot shall consist of all solutions or chemicals of one class for delivery at one time.

4.1.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.1.2 and 6.2).

4.1.3 Sampling for tests. Sampling of a lot for test purposes shall be as specified in table II.

4.1.4 Examination of filled containers. Each filled container in the sample selected as specified in 4.1.3 shall be examined to verify compliance with all requirements of this specification regarding fill, marking, and other requirements not involving tests. Containers shall be examined for defects of the container, including leakage. Each sample-filled container shall also be weighed to determine the amount of the contents (see 4.2.1). Any container in the sample having one or more defects or containing less than the required amount of chemical shall be cause for rejection of that container.

4.1.5 Test glassware, reagents, and conditions. Volumetric glassware used in the test procedures (see 4.2) shall be class A glassware in accordance with ASTM E287, ASTM E288, or ASTM E969, as applicable. Water used in the test procedures shall be in accordance with ASTM D1193, type II reagent water. Test solutions shall be prepared from reagent grade chemicals in accordance with ACS Reagent Chemicals: Specifications and Procedures for Reagents and Standard-Grade Reference Materials. Titrations shall be conducted at 20 to 25 ºC.

4.1.6 Rejection of lots. If a sample fails to pass any of the tests specified in table II, the production lot represented by that sample shall be rejected. As specified (see 6.2), delivery of the material shall be withheld pending notification of the results of tests.

4.1.7 Test reports. When specified (see 6.2), test reports shall be prepared.

4.2 Test procedures.

4.2.1 Weight (classes E, I, K, and P). Prior to performing applicable chemical tests (see 4.2.1, 4.2.2, 4.2.3, or 4.2.4), any sealing band or tape shall be removed from the sample bottles, sample identification numbers shall be marked on the bottles with an indelible marker, and then each sample-filled bottle, including its cap, shall be weighed. After performing the applicable chemical tests, any remaining liquid in the bottle shall be poured out, the cap shall be replaced, and each empty bottle shall be weighed. Sample weight shall be calculated using the following equation:

$$Weight\ (grams) = W_f - W_e$$

Where:

W_f = Weight of the sample-filled bottle (grams)

W_e = Weight of the emptied bottle (grams)

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4.2.2 Class E – standard hardness solution, 0.20 epm hardness. The total hardness concentration shall be measured in accordance with the ASTM D1126, 9.2 low hardness method. Reagents in accordance with MIL-DTL-24617 may be substituted for the reagents specified in ASTM D1126.

4.2.3 Class I – standard conductivity solution, 1400 µS/cm. The cell constant of the conductivity cell that will be used for testing the standard conductivity solution shall be determined in accordance with ASTM D1125, method A. The electrical conductivity of the standard conductivity solution shall be measured in accordance with ASTM D1125, method A.

4.2.4 Class K – standard chloride solution, 2.00 epm chloride. The chloride ion concentration shall be measured in accordance with ASTM D512, method A (mercurimetric, titration). A chloride indicator in accordance with MIL-DTL-15000 or appendix A may be used instead of the "mixed indicator solution". The chloride ion concentration shall be calculated in epm using the following formula:

$$Chloride\ (epm) = \frac{[(V_1 - V_2) \times N]}{S}$$

Where:

V_1 = standard $Hg(NO_3)_2$ solution required for titration of the sample, milliliters (mL)

V_2 = standard $Hg(NO_3)_2$ solution required for titration of the blank, mL

N = normality of the standardized $Hg(NO_3)_2$ solution

S = volume of the sample used in the titration, mL

4.2.5 Class P – standard methyl purple alkalinity (6.0 epm).

4.2.5.1 Preparation of reagents used in the test.

4.2.5.1.1 Sodium carbonate standard. 0.050 N sodium carbonate (2.650 grams Na_2CO_3 per liter) shall be prepared from ACS reagent grade sodium carbonate that has been dried at 250 ºC for a minimum of 4 hours and then cooled in a desiccator. The sodium carbonate shall be dried within 24 hours before preparing the standard.

4.2.5.1.2 Methyl purple indicator. The methyl purple indicator shall be in accordance with the requirements in appendix A. Methyl red without a blue dye, methyl orange, and methyl violet shall not be considered acceptable substitutes in this application.

4.2.5.1.3 Nitric acid reagent. Nominal 0.05 N nitric acid shall be prepared and standardized against 20 mL of 0.050 N sodium carbonate (see 4.2.5.1.1), using 2 to 3 drops of methyl purple indicator (see 4.2.5.1.2), and titrating until the final purple endpoint. The sodium carbonate solution shall not be boiled. The standardized normality of the nitric acid shall be calculated as the average of three determination using the following formula:

$$Standardized\ normality\ (N_N)\ of\ nitric\ acid = \frac{0.050 \times V_C}{V_N}$$

Where:

V_C = Volume of sodium carbonate solution (20 mL)

V_N = Volume of nitric acid required for titration of the blank, mL

4.2.5.2 Alkalinity test. The alkalinity test shall be performed as follows:

a. Pipet 50 mL of the standard into a clean casserole.

b. Add 2 to 3 drops of methyl purple indicator (see 4.2.5.1.2).

c. Titrate carefully with standardized nitric acid (see 4.2.5.1.3) until the final purple endpoint is obtained.

d. Calculate the alkalinity of the sample as follows:

$$Alkalinity, epm = \frac{V_N \times N_N}{0.0500}$$

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Where:

V_N = Volume of standardized nitric acid required for titration of the sample, mL

N_N = Standardized normality of nitric acid (see 4.2.5.1.3)

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reagent standards covered by this specification are intended to be used in testing shipboard boiler water and feedwater in accordance with the procedure requirements of S9086-GX-STM-020/220, Boiler Water/Feedwater Test and Treatment. These chemicals are used in conjunction with MIL-W-16642 and MIL-DTL-15000. Current stock of materials may be used until depleted; future requisitions for materials should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition requirements should specify the following:

a. Title, number, and date of this specification.
b. Class or classes of chemicals required (see 1.2).
c. Requirement for prohibited materials verification (see 3.1.2 and 4.1.2).
d. Test samples required (see 4.1.3).
e. Requirements for delayed delivery of the material pending results of tests (see 4.1.6).
f. When test reports are required (see 4.1.7).
g. Packaging and marking requirements (see 5.1, 6.3, and 6.4).
h. Safety Data Sheets (SDS), when required (see 6.5).

6.3 Suggested marking. In addition to any special marking specified (see 6.2), unit, interior, and exterior containers should be marked in accordance with MIL-STD-129 and should include the following:

a. National stock number (NSN).
b. Name and class of item.
c. Manufacturer's or contractor's name, or both.
d. Contract number.
e. Quantity/net content.
f. Date of manufacture, month, and year.
g. Expiration date of 60 months from date of manufacture.
h. "Note: Mandatory shelf-life material. Discard material after expiration date."
i. Lot or batch number.

6.4 Hazardous chemicals. As specified (see 6.2), each package unit of hazardous chemicals should have affixed thereto warning labels in accordance with Code of Federal Regulations (CFR) 49 CFR 171-199, 29 CFR 1910.1200, or appropriate Department of Defense instructions.

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6.5 Safety data sheets (SDS). When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.6 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-Life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.7 NSNs. Table III provides a list of assigned NSNs that correspond to this specification. This list may not include all possible NSNs associated with this document.

TABLE III. List of applicable NSNs.

Class	Description	NSN	Shelf Life Code (SLC)	Unit of Issue
E	Standard hardness solution (0.2 epm)	6810-01-125-5234	S 1/	BT 2/
I	Standard conductivity solution (1400 µS/cm)	6810-00-945-7682	S	BT
K	Standard chloride solution (2.0 epm)	6810-01-129-3762	S	BT
P	Standard alkalinity solution (6.0 epm)	6810-01-259-8231	S	BX 3/

NOTES:

1/ S: 5 years (60 months)

2/ BT: Bottle

3/ BX: Box (contains 6 bottles)

6.8 Subject term (key word) listing.

Alkalinity

Chloride

Conductivity

Hardness

Mercuric nitrate

Methyl purple

Nitric acid

Sodium carbonate

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TITRATION INDICATORS

A.1 SCOPE

This appendix covers the composition and preparation of indicators used in titration tests for standard alkalinity solutions (see 4.2.5.1.2) and standard chloride solutions (see 4.2.4). This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in this appendix. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-15000 - Water-Testing Chemicals, Boiler, Shipboard Use

(Copies of this document are available online at https://quicksearch.dla.mil.)

A.2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

UNITED STATES PATENT AND TRADEMARK OFFICE

Patent No. 2,416,619 - Chemical Indicator

Patent No. 2,784,064 - Indicator for Chloride Titrations

(Copies of these documents are available online at www.uspto.gov/.)

A.2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D512 - Standard Test Methods for Chloride Ion in Water

(Copies of this document are available online at www.astm.org.)

SOCIETY OF DYERS AND COLOURISTS (SDC)

Colour Index

(Copies of this document are available online at http://colour-index.com.)

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A.3 TITRATION INDICATORS USED IN VERIFICATION TESTS

A.3.1 Methyl purple indicator.

A.3.1.1 Description. The methyl purple indicator used to detect the titration end point in the test for class P – standard methyl purple alkalinity (6.0 epm) (see 4.2.5.1.2) shall consist of a mixture of methyl red sodium salt (Chemical Abstracts Service [CAS] Registry Number 845-10-3) and a blue dye dissolved in reagent grade water or other solvent such that the pH of the titrand will be green at pH 5.5, have a grey or clear transition phase, and will change to purple at a pH of 4.8±0.2. This indicator was originally described by now-expired U.S. Patent No. 2,416,619.

A.3.1.2 Preparation. Table A-I provides a list of some blue dyes that have been successfully used to prepare the methyl purple indicator, including the CAS registry number and a cross reference to the Colour Index. U.S. Patent No. 2,416,619 provides specific preparation instructions for several of these dyes.

TABLE A-I. Blue dyes that have been successfully used to prepare methyl purple indicator.

Blue Dye	CAS Registry Number	Color Index Number	
		1st Edition 1/	2nd Edition 2/
Acid Blue 5	3374-30-9	714	42,052
Acid Blue 7	3486-30-4	673	42,080
Acid Blue 22	28631-66-5	707	42,755
Acid Blue 102	6856-08-2	833	50,320
Direct Blue 1	2610-05-1	518	24,410

NOTES:

1/ Colour Index, Society of Dyers and Colourists, Yorkshire England, 1st ed., 1924

2/ Colour Index, Society of Dyers and Colourists, Yorkshire England, 2nd ed., 1957, and later editions

A.3.1.3 Unacceptable substitutes. Methyl red without a blue dye, methyl orange, and methyl violet, shall not be considered acceptable substitutes in this application.

A.3.2 Chloride indicator.

A.3.2.1 Description. If ASTM D512 is used to determine the chloride content in class K – standard chloride solution (see 4.2.4), then a mixture of diphenylcarbazone (CAS Registry Number 538-62-5) and bromophenol blue (CAS Registry Number 115-39-9), dissolved in ACS reagent grade ethanol, methanol, or isopropanol, prepared using the instructions provided by now-expired U.S. Patent No. 2,784,064, may be used instead of the “mixed indicator solution” in accordance with ASTM D512. Chloride indicator in accordance with MIL-DTL-15000 class F may also be used.

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CONCLUDING MATERIAL

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Review activities:
Navy – OS
DLA – GS

Preparing activity:
Navy – SH
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